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Riccardo Tedoldi

Computer Scientist & Enthusiast Deep Learning scholar and researcher
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Education

- **University of Trento** — Trento, TN
  M. Sc.; Artificial Intelligence Systems (Current GPA: 3.92/4) - Full Courses List — *Sep 2022 - Present*
- **University of Parma** — Parma, PR
  B. Sc.; Computer Science GPA: 3.854/4 - First graduated in class of 150 *- Full Courses List* — *Oct 2019 - Jul 2022*
- **Boarding school Maria Luigia** — Parma, PR
  Scietific high-school cerfificate; — *Jun 2019*
  Subjects: Math, Physics, Chemistry, Biology, Philosophy, History, Latin, English literature, Italian literature, and Art.

Experience

- **Research Intern** — Parma, PR
  Bioinformatics & Molecular Biology Intern — *Oct 2021 - Jun 2022*
  - **Clustering Algorithm**: Developed a clustering algorithm that aimed to identify relevant atoms important for the protein binding. Supervisior: Dr. Marco Malatesta
    The algorithm aimed to reduce the computational complexity of the previous one from $\mathcal{O}(n^2)$ to $\mathcal{O}(n)$ with comparable/better performance.
  - **Neo-functionalization detection**: Developed Deep Learning Transformers-based architectures (i.e. BERT, ViT, CCT, MaXViT) to detect whether mutations change the protein function. We worked with different protein representations (i.e. sequence, structure, embedding and more). Supervisiors: PhD. Carlo DeRito, Dr. Marco Malatesta, Prof. Eleonora Iotti

Research

- **Decoding the hidden Waves: Revealing the nature of brain activity** — Trento, IT
  Seminar Paper, ***R. Tedoldi****; Supervisor: U. Hasson; Code: (github)* — *Jul 2022*
- **An open source perspective on AI and alignment with the EU AI Act** — Macao, CHN
  IJCAI AISafety 2023, Author: D. Calanzone, A. Coppari, ***R. Tedoldi****, G. Olivato, C. Casonato* — *Jun 2022*
- **A Transformer-based Machine Learning Approach to Detect Protein Neofunctionalization** — Parma, IT
  B.Sc. Thesis, ***R. Tedoldi****; Supervisor: E. Iotti; Co-supervisors: C. De Rito, A. DalPalù* — *Jun 2022*

Research Projects

- **Text-to-3D**: Conditional generation through diffusion model (github); *Condition the generation with multiple-views we state better divergence/chamfer distance scores (on ModelNet40 and ShapeNet).*
- **D2BL**: Dynamic Distribution-driven Backpropagation Learning (github); *Developed a mechanism to recombine the hidden features of the network. In a simple MLP with 4 layers, we improved the accuracy (+12%) by substituting the last layer with our mechanism.*
- **ShadowMe**: Low-power 2D Motion Emulation and 3D Renders Learning (github); *Developed a low-power pipeline with a web interface to emulate human motion in real-time/mp4. It works on iPad for* 25/30*fps.*
- **UHC**: A study on homologous protein latent encoding (github); *Relying on* 15 *human sequences we collected the best matching sequences in other species to identify similarities in their function.*
- **Visual Grounding**: Visual Grounding on RefCOCOg (github); *Designed* 2 *solutions that achieved* +4% *AP compared to the baseline solution for the Deep Learning Course.*

Skills Summary

- **Languages**: Python, C/C++, SQL, Bash, LaTeX, Unix Terminal, Prolog **Basic**: CUDA, Java, R, Javascript
- **Tools**: PyTorch, PyTorch-Lightning, PyTorch3D, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, MNE, OpenCV, Git, Docker, Linux, Windows, MacOS, Jupyter, Colab, Office, Adobe Illustrator, Adobe Lightroom

Honors and Awards

- Grant on the tuition fees of 300€: The grant is awarded to the top 11 students of the Department of Computer Science of the University of Parma. I was awarded the grant in both consecutive years of my Bachelor's degree (2020/2021, 2021/2022).
- First B.Sc. Degree of the Class, Computer Science Jul 2022

Extra Curricula

- **Volunteer for the homeless**: The activity consists in distributing food and drinks to the homeless (Sant'Egidio Community)
- **Summer entertainer volunteer**: The activity consists in organizing games and activities for children and teenagers.